UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523

DATABASE WBL

SESSION 2023/2024 – SEMESTER 1

ASSIGNMENT 1

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YEAR/COURSE: 2/SECPH

SECTION 2

QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	**Order**	**Customer**
Information	**Iphone16**	**Name**

QUESTION 2

Answer the questions below based on the following relational schema:

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)

EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)

PROJECT (Proj_Code, Proj_Name, Proj_StartDate)

SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE (entit y)	PRIMARY KEY	FOREIGN KEY
SCIENTIST	**Sc ID**	**-**
EXPERIMENT	**Exp No**	**Proj Code**
PROJECT	**Proj Code**	**-**
SCIENTIST_EXPERIMENT	**Time**	**Sc_ID, Exp NO**

b) Does the table exhibit entity integrity? Answer **'Yes'** or **'No'** and then explain your answer. (4M)

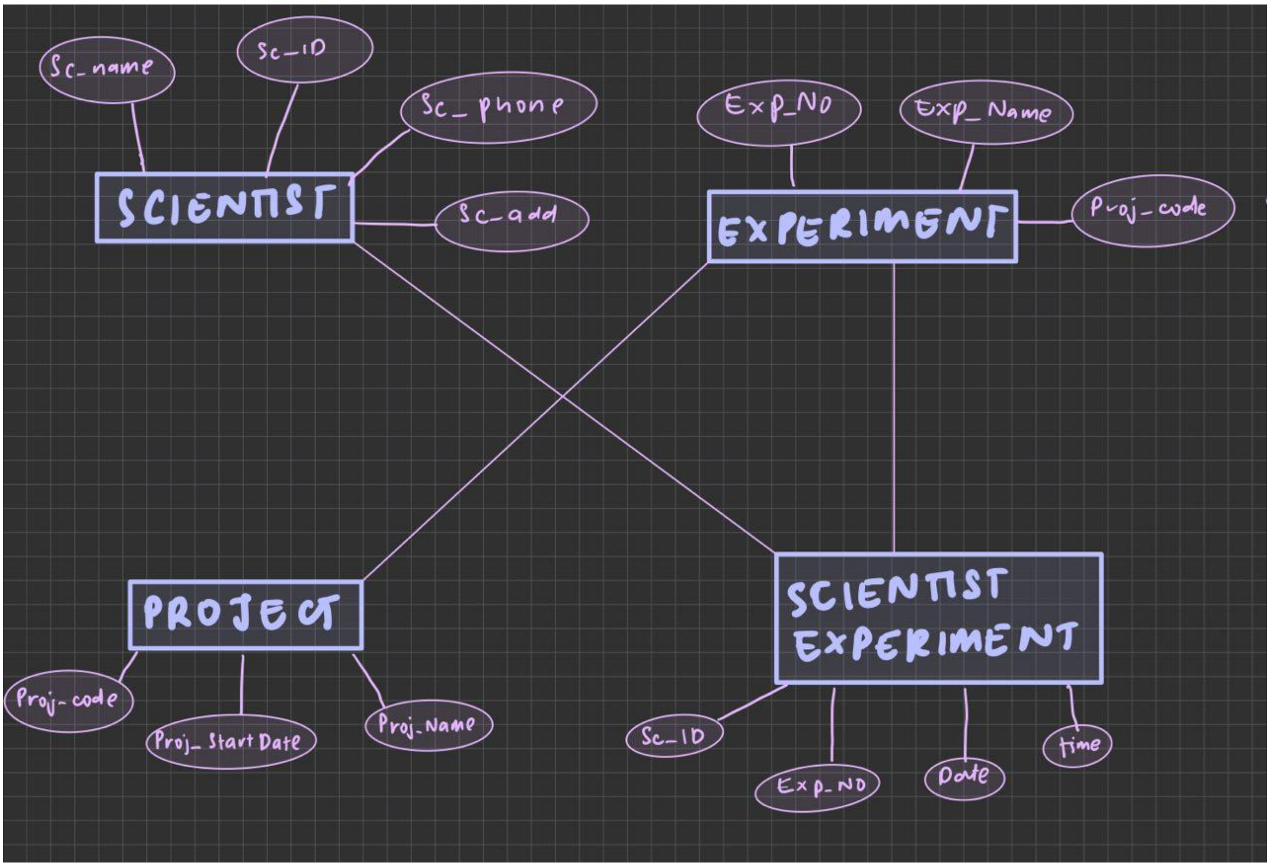
TABLE	ENTITY INTEGRITY (entity that has a primary key)	EXPLANATION
SCIENTIST	**Yes**	**Because it has a primary key**
EXPERIMENT	**Yes**	**Because it has a primary key**
PROJECT	**Yes**	**Because it has a primary key**

SCIENTIST_EXPERIMENT	**Yes**	**It has a primary key**

c) Does the table exhibit referential integrity? Answer **'Yes'** or '**No'** and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY (entities with foreign key)	EXPLANATION
SCIENTIST	**No**	**No foreign key**
EXPERIMENT	**Yes**	**It has a foreign key (Proj code)**
PROJECT	**No**	**No foreign key**
SCIENTIST_EXPERIMENT	**Yes**	**It has foreign keys (Sc ID and Exp No**

d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

Task:

1. List each Entity that is identified in Luxury Records System.
   Musician, instrument, album, song.

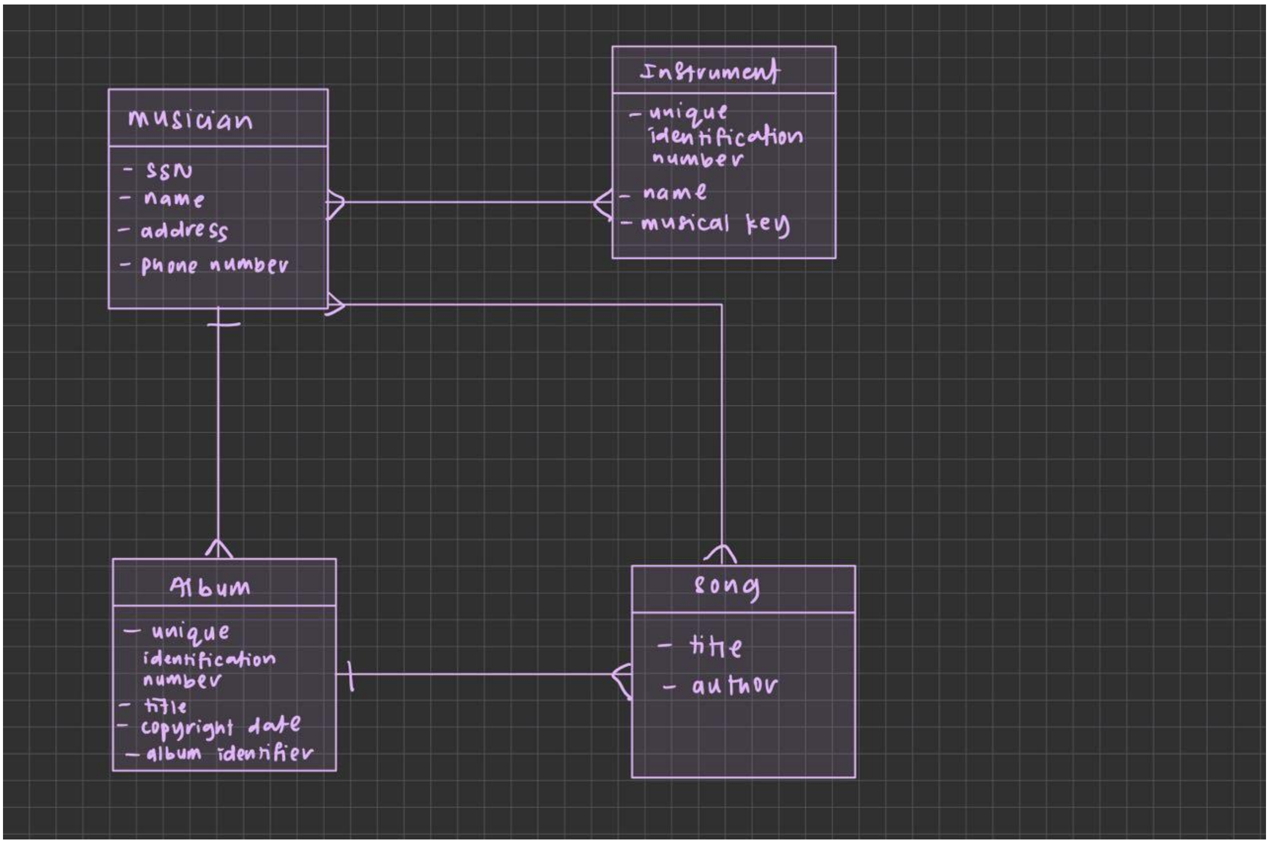
2. For each entity list the attributes.
   Musician - an SSN, a name, an address, and a phone number

   Instrument - unique identification number, a name(e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).

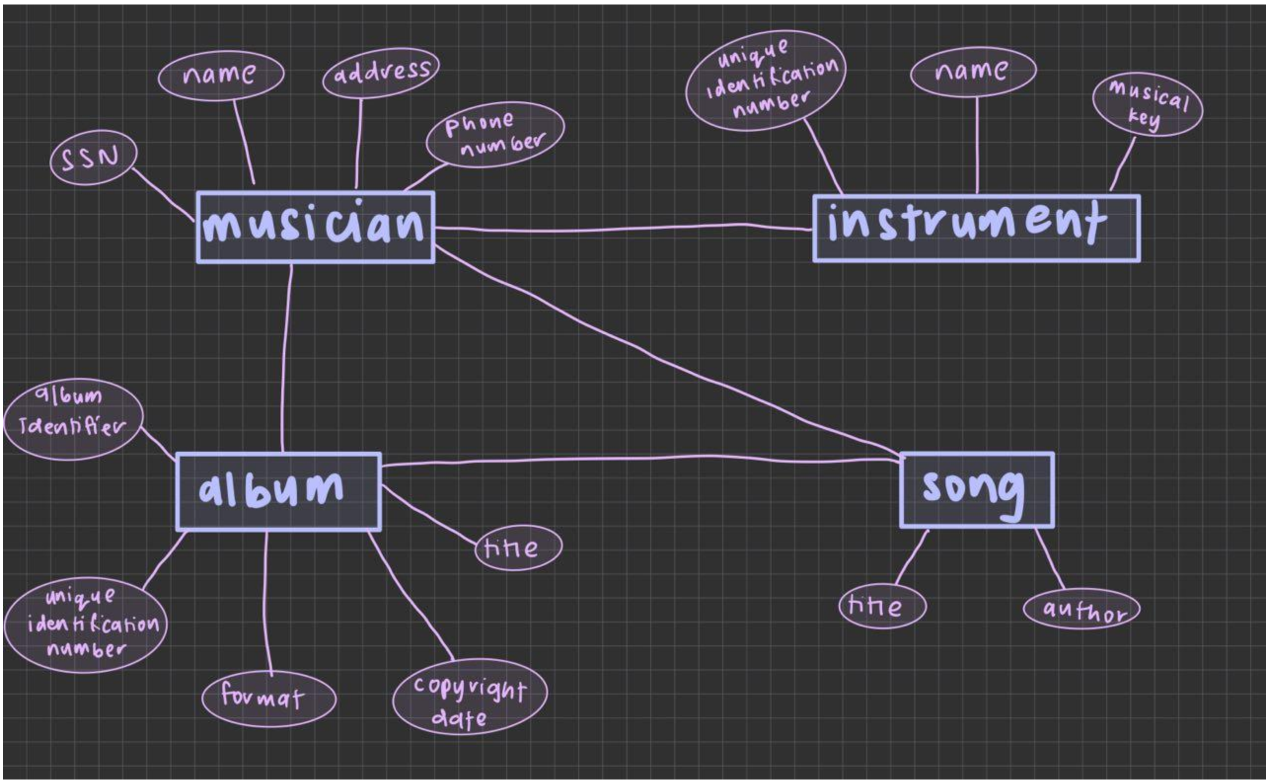
   Album - unique identification number, a title, acopyright date, a format (e.g., CD or MC) and an album identifier.

   Song - title and an author

3. Draw a logical model to represent each entity and the relationships between them.



4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.



QUESTION 4

Give an Give an example of each of the three types of relationships.

i) One-to-many relationship

Man and cars

ii) Many-to-many relationship
Customers and products

iii) One-to-one relationship
A student and one uni program

iv) Many-to-one relationship
Cars and man

QUESTION 5

CLIENT (Client_Number, Clent_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

a) What is the relationship between Recruiter and Client?
Many to one (1 client has many recruiters)

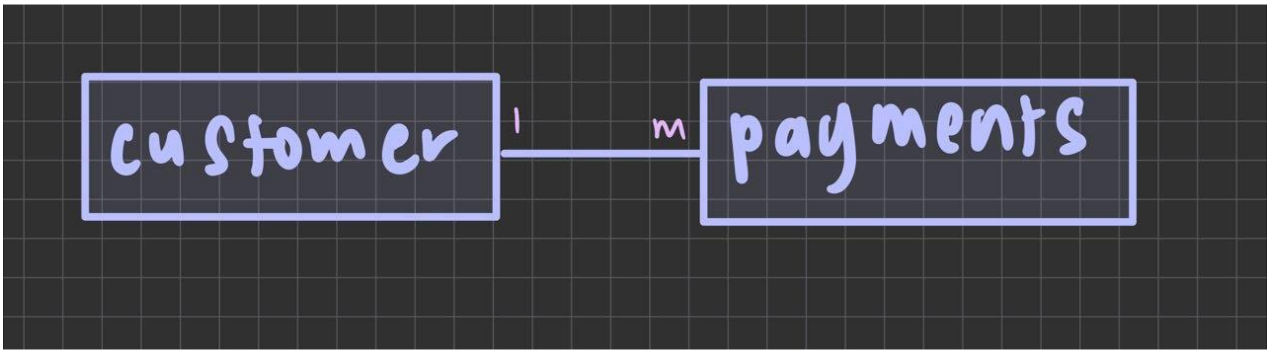
b) What is the primary key for the Client table and Recruiter table?
Client_Number, Recruiter_Number

c) Define foreign key? Identify foreign key in the Client table.
A column in a table that links to a primary key in another table

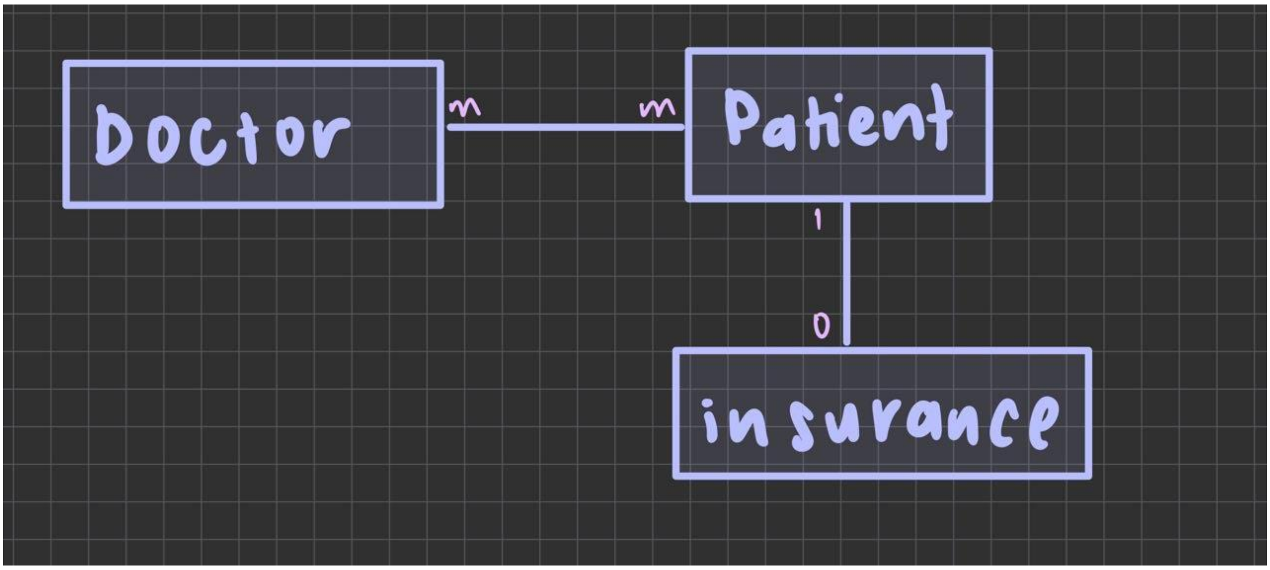
QUESTION 6

Draw the Chen & Crow's Foot notation based on the following statements:

i) "A customer can make many payments, but each payment is made by only one customer."



ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book_ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.